



Seeing the detail

The light-sensitive part of the eye is the retina, which lines the back of the eye. It is crammed with receptor cells—some rod-shaped, others cone-shaped. When stimulated by light, these send electrical nerve impulses to the brain. While the rods can work in dim light, cones need brighter light, but they help the animal see things in more detail and in color.

Fovea

The fovea is a concentrated spot of cone cells on the retina, which helps the owl pick out lots of fine detail.

Vitreous humor

Behind the lens the eye is filled with a jelly called vitreous humor, which helps the eye maintain its shape.

Sclerotic ring

A ring of bone surrounds the eye and helps to keep it firmly in position.

Pecten

A comblike structure of blood vessels (not found in humans) helps to nourish the eye.

Retina

Muscle

The muscles that move the eye are not as well developed in birds as they are in humans.

Lens

A large lens bends light rays to focus them on the retina.

Cornea

Light rays bend slightly when they enter the eye through the transparent cornea.

Aqueous humor

Liquid between the cornea and lens is called aqueous humor.

Ciliary muscles

Connective tissue joins the lens to ciliary muscles, which help to change the lens shape to alter focus from near and far objects.

Layers of the eye

As well as the light-sensitive retina, the eye has two other layers: the sclera, and the choroid. The sclera is the tough outer layer—in humans extending around the front to form the “white” of the eye. The choroid is packed with blood vessels and provides the eye’s oxygen supply. It also contains a dark pigment, which in day-active animals stops light from being reflected too much inside the eye.

Iris and pupil

The iris is located at the front of the eye, just behind the transparent cornea and a layer of clear liquid. It forms a bright colored ring with a dark hole at its center—the pupil—which is where light enters. Iris muscles control the amount of light coming into the eye by expanding the pupil in dim light and making it shrink in bright light.

